Chankyu (Charlie) Han

https://chankyu-charlie.github.io/ | chankyu@uw.edu

Research Interests

Smart Sensing and Haptic with AI/ML, Tactile Sensing, Human-Computer Interaction, Fabrication

Education

University of Washington (UW) — Seattle, Washington

Ph.D. in Electrical and Computer Engineering, Advisor: Dr. Yiyue Luo — Jun.2025 - Present

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, Korea

M.S. in Mechanical Engineering, Advisor: Dr. Inkyu Park — Mar.2021 - Feb.2023

B.S. in Mechanical Engineering — Sep.2016 - Feb.2021
Graduation with *Cum Laude*

Publications

(†: equal contribution)

[11] MagBall: Magnetic Rollerball for Multi-Scale Contact Interactions on Diverse Surfaces

C. Han, Y. Miao, S. Zhang, Y. Luo

CHI 2026: ACM Conference on Human Factors in Computing Systems (accepted)

[10] Thermoforming 2D Films into 3D Electronics for Customizable Tactile Sensing

J. Choi†, **C. Han**†, D. Lee†, H. Kim, G. Lee, J.-H. Ha, Y. Jeong, J. Ahn, H. Park, H. Han, S. Cho, J. Gu, I. Park

Science Advances 11.20 (2025): eadv0057. *https://doi.org/10.1126/sciadv.adv0057*

[9] Spike-based Self-calibration for Enhanced Accuracy in Self-powered Pressure Sensing

C. Han†, J. Choi†, J. Ahn, H. Kim, J.H. Ha, H. Han, S. Cho, Y. Jeong, J. Gu, I. Park

Advanced Materials Technologies (2023): 2301199. *https://doi.org/10.1002/admt.202301199*

[8] Recent Advances in Sensor–Actuator Hybrid Soft Systems: Core Advantages, Intelligent Applications, and Future Perspectives

C. Han†, Y. Jeong†, J. Ahn†, T. Kim, J. Choi, J.-H. Ha, H. Kim, S.H. Hwang, S. Jeon, J. Ahn, J.T. Hong, J.J. Kim, J.H. Jeong, I. Park

Advanced Science (2023): 2302775. *https://doi.org/10.1002/advs.202302775*

[7] Nanotransfer Printing of Functional Nanomaterials on Electrospun Fibers for Wearable Healthcare Applications

J.-H. Ha, J. Ko, J. Ahn, Y. Jeong, J. Ahn, S. Hwang, S. Jeon, D. Kim, S. A. Park, J. Gu, J. Choi, H. Han, **C. Han**, B. Kang, B.-H. Kang, S. Cho, Y.J. Kwon, C. Kim, S. Choi, G.-D. Sim, J.-H. Jeong, I. Park

Advanced Functional Materials (2024): 2401404. *https://doi.org/10.1002/adfm.202401404*

[6] Wireless, Multimodal Sensors for Continuous Measurement of Pressure, Temperature, and Hydration of Patients in Wheelchair

S. Cho, H. Han, H. Park, S.-U. Lee, J.-H. Kim, S.W. Jeon, M. Wang, R. Avila, Z. Xie, K. Ko, M. Park, J. Lee, M. Choi, J.-S. Lee, W.G. Min, B.-J. Lee, S. Lee, J. Choi, J. Gu, J. Park, M.S. Kim, J. Ahn, O. Gul,

C. Han, K. Lee, S. Kim, K. Kim, J. Kim, C.-M. Kang, J. Koo, S.S. Kwak, S. Kim, D.Y. Choi, S. Jeon, H.J. Sung, Y.B. Park, Y.T. Choi, M. Je, Y.S. Oh, I. Park

npj Flexible Electronics 7.1 (2023): 8. https://doi.org/10.1038/s41528-023-00238-3

[5] Battery‐Free, Wireless, Ionic Liquid Sensor Arrays to Monitor Pressure and Temperature of Patients in Bed and Wheelchair

H. Han, Y.S. Oh, S. Cho, H. Park, S.-U. Lee, K. Ko, J.-M. Park, J. Choi, J.-H. Ha, **C. Han**, Z. Zhao, Z. Liu, Z. Xie, J.-S. Lee, W.G. Min, B.-J. Lee, J. Koo, D.Y. Choi, M. Je, J.-Y. Sun, I. Park

Small 19.9 (2023): 2205048. https://doi.org/10.1002/smll.202205048

[4] All-Recyclable Triboelectric Nanogenerator for Sustainable Ocean Monitoring Systems

J. Ahn† , J.-S. Kim† , Y. Jeong, S. Hwang, H. Yoo, Y. Jeong, J. Gu, M. Mahato, J. Ko, S. Jeong, J.-H. Ha, H.-S. Seo, J. Choi, M. Kang, **C. Han**, Y. Cho, C. H. Lee, J.-H. Jeong, I.-K. Oh, I. Park

Advanced Energy Materials (2022): 2201341. https://doi.org/10.1002/aenm.202201341

[3] A Review of Recent Advances in Electrically-Driven Polymer-based Flexible Actuators: Smart Materials, Structures, and Their Application

J. Ahn† , J. Gu† , J. Choi† , **C. Han**, Y. Jeong, J. Park, S. Cho, Y. Oh, J.-H. Jeong, M. Amjadi, I. Park

Advanced Materials Technologies (2022): 2200041. https://doi.org/10.1002/admt.202200041

[2] Stretchable Printed Circuit Board Based on Leak-Free Liquid Metal Interconnection and Local Strain Control

M. S. Kim, S. Kim, J. Choi, S. Kim, **C. Han**, Y. Lee, Y. Jung, J. Park, S. Oh, B.-S. Bae, H. Lim, I. Park

ACS Applied Materials & Interfaces 14.1 (2021): 1826-1837. https://doi.org/10.1021/acsami.1c16177

[1] Customizable, Conformal, and Stretchable 3D Electronics via Pre-distorted Pattern Generation and Thermoforming

J. Choi, **C. Han**, S. Cho, K. Kim, J. Ahn, D. D. Orbe, I. Cho, Z. Zhao, Y.Oh, H. Hong, S. S. Kim, I.Park

Science Advances 7.42 (2021): eabj0694. https://doi.org/10.1126/sciadv.abj0694

Technology Transfer

Joint transfer of the following technologies ($37,000, 2022):

Finger wearable touch sensor, control system using same, and method thereof

I. Park† , J. Choi† , **C. Han**† , G. Lee†

Korea Patent (Application Number: 10-2021-0096964), 2021.

Large-Area Multi-Pressure Sensor Pad Fabrication Technology (Know-how)

I. Park† , J. Choi† , **C. Han**† , G. Lee†

Research Experiences

UW Wearable Intelligence Lab Jun.2025 - Present
Research Assistant

Mentor: Yiyue Luo

Korea University Sensor and Nanofabrication System Lab Jan.2025 – Jun.2025
Visiting Researcher

Mentors: Junseong Ahn, Jungrak Choi

KAIST Machine Intelligence and Novel-sensor Technology Lab Jun.2020 – Jun.2023
Undergraduate Intern & Research Assistant

Mentor: Inkyu Park

Scholarships & Awards

Best Oral Presentation Award May 2023
IEEE NEMS 2023 [9]

Best Poster Award Nov. 2021
International Conference on Advanced Electromaterials (ICAE) 2021 [1]

National Excellence Scholarship (Graduate Program) Mar.2021 – Feb.2023
Government funding for admission fees, full amount of school support fees in the master's program

KAIST Undergraduate Research Grant Sept. 2020 - Feb. 2021
Title: Customizable, Conformal, and Stretchable 3D Electronics via Pre-distorted Pattern Generation and Thermoforming [1]

Dean's List, School of Freshman Feb. 2017
Top 2% of KAIST freshman cohort

National Excellence Scholarship (Undergraduate Program) Sep.2016 – Feb.2020
Government funding for admission fees, full amount of school support fees in the undergraduate program

Service

Academic Service

Student Volunteer: CHI 2026

Reviewer

Nature Communications (1), CHI 2026

Outreach and Leadership

Workshop Organizer: Everyday Wearable for Personalized Health and Well-Being CHI 2026

High School Outreach: Talk and Lab Tour on Wearable Intelligence 2025

Skills

Languages: Python, C/C++, MATLAB, LabVIEW

Fabrication: 3D printing (DLP, FDM), Thermoforming, Laser processing(CO_2, UV, Fiber), PCB design

TOEFL: 108/120

Others

Military Service in Republic of Korea Army Jun.2023 – Dec.2024